



Lab Assignment

Course: Higher Level Programming Language Lab

Course Code: CSEL-3204

Submitted To

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1. Write a program to remove duplicate values from a list and display only unique elements.

```
data = [10,20,20,30,40,40]
unique = []

for num in data:
    if num not in unique:
        unique.append(num)
print("Unique numbers: ", unique)
```

Output: Unique numbers: [10, 20, 30, 40]

2. Write a program to find the maximum and minimum values in a list.

```
list = [26,15,38,46,29]
maxi = max(list)
mini = min(list)
print("Maximum number: ",maxi)
print("Minimum number: ",mini)
```

Output: Maximum number: 46

Minimum number: 15

3. Write a program to sort a list in ascending and descending order.

```
list = [63,27,19,17,40,29]

list.sort()
print("Ascending list: ",list)

list.sort(reverse = True)
print("Descending list: ",list)
```

Output: Ascending list: [17, 19, 27, 29, 40, 63]

Descending list: [63, 40, 29, 27, 19, 17]

4. Write a program to modify an element of a tuple by converting it into a list and then back to a tuple

```
tup = (10,20,49,73)
```

```
li = list(tup)
```

```
li[2] = 99
```

```
mod_tup = tuple(li)
```

```
print('Original tuple: ',tup)
```

```
print('Modified tuple: ',mod_tup)
```

Output: Original tuple: (10, 20, 49, 73)

Modified tuple: (10, 20, 99, 73)

5. Write a program to perform union, intersection, and difference on two sets.

```
set1 = {1,2,3,4}
```

```
set2 = {3,4,5,6}
```

```
un = set1.union(set2)
```

```
inte = set1.intersection(set2)
```

```
diff = set1.difference(set2)
```

```
print('Union of two sets: ',un)
```

```
print('Intersection of two sets: ',inte)
```

```
print('Difference of two sets: ',diff)
```

Output: Union of two sets: {1, 2, 3, 4, 5, 6}

Intersection of two sets: {3, 4}

Difference of two sets: {1, 2}

6. Write a program to check whether a given element exists in a set.

```
my_set = {10, 20, 30, 40}
```

```
element = int(input("Enter element to check: "))
```

```
if element in my_set:
```

```
    print("Element exists in the set")
```

```
else:
```

```
    print("Element does not exist in the set")
```

Output: Enter element to check: 20

Element exists in the set

7. Write a program to create a dictionary of student marks, add a new student, update marks, and find the student with the highest marks.

```
students = {'A': 80, 'B': 75, 'C': 90}
```

```
students['D'] = 85
```

```
students['B'] = 82
```

```
max_student = max(students, key=students.get)
```

```
max_marks = students[max_student]
```

```
print("Updated Dictionary:", students)
```

```
print("Highest Marks:", max_student, "=", max_marks)
```

Output: Updated Dictionary: {'A': 80, 'B': 82, 'C': 90, 'D': 85}

Highest Marks: C = 90

8. Write a program to count how many times each element appears in a list.

```
data = [1,2,2,3,3,3]
```

```
dict = {}
```

```
for num in data:
```

```
    if num in dict:
```

```
        dict[num] += 1
```

```
    else:
```

```
        dict[num] = 1
```

```
print('Number of each element: ',dict)
```

Output: Number of each element: {1: 1, 2: 2, 3: 3}

9. Write a program to find common elements between two sets.

```
set1 = {10,20,30,40}
```

```
set2 = {30,40,50,60}
```

```
set3 = set1.intersection(set2)
```

```
print('The common elements in a set: ',set3)
```

Output: The common elements in a set: {40, 30}

10. Write a program to create a dictionary using two lists (keys and values).

```
keys = ['a', 'b', 'c']
```

```
values = [1, 2, 3]
```

```
result = dict(zip(keys, values))
```

```
print(result)
```

Output: {'a': 1, 'b': 2, 'c': 3}

11. Write a program to count the frequency of each word in a sentence and display unique words.

```
print("Unique Words:", unique_words) sentence = 'apple
banana apple orange banana apple'

words = sentence.split()

frequency = {}

for word in words:
    if word in frequency:
        frequency[word] += 1
    else:
        frequency[word] = 1

unique_words = set(words)
print("Word Frequency:", frequency)
```

Output:

Word Frequency: {'apple': 3, 'banana': 2, 'orange': 1}

Unique Words: {'apple', 'banana', 'orange'}

Advanced 1: The FizzBuzz list

```
fb_list = []

for i in range(1, 16):
    if i % 3 == 0 and i % 5 == 0:
        fb_list.append('FizzBuzz')
    elif i % 3 == 0:
        fb_list.append('Fizz')
    elif i % 5 == 0:
        fb_list.append('Buzz')
    else:
```

```
        fb_list.append(i)
print(fb_list)
```

Output: [1, 2, 'Fizz', 4, 'Buzz', 'Fizz', 7, 8, 'Fizz', 'Buzz', 11, 'Fizz', 13, 14, 'FizzBuzz']

2. Removing negative values (Filtering)

```
temp = [22, -5, 18, -1, 30, 25, -10]
newTemp = []
```

```
for num in temp:
    if num>=0:
        newTemp.append(num)
print('The original list: ',temp)
print('The modified list: ',newTemp)
```

Output: The original list: [22, -5, 18, -1, 30, 25, -10]

The modified list: [22, 18, 30, 25]

3. E-Commerce Analytics

```
sales_data = [
    (101, "Laptop", "Electronics", 1200),
    (102, "Mouse", "Electronics", 25),
    (101, "Laptop", "Electronics", 1200),
    (103, "Monitor", "Electronics", 300),
    (104, "Desk Chair", "Furniture", 150),
    (105, "Lamp", "Furniture", 45),
    (106, "Mouse", "Electronics", 25)
]
```

```
unique_transactions = {}
```

```

for tid, product, category, price in sales_data:
    if tid not in unique_transactions:
        unique_transactions[tid] = (product, category,
price)

category_products = {}
for product, category, price in
unique_transactions.values():
    if category not in category_products:
        category_products[category] = set()
    category_products[category].add(product)

revenue = {}
for product, category, price in
unique_transactions.values():
    if product in revenue:
        revenue[product] += price
    else:
        revenue[product] = price

max_price_per_category = {}

for product, category, price in
unique_transactions.values():
    if category not in max_price_per_category:
        max_price_per_category[category] = (product, price)
    else:
        if price > max_price_per_category[category][1]:
            max_price_per_category[category] = (product,
price)

```



```
print("--- Unique Products per Category ---")

for category, products in category_products.items():
    print(f"{category}: {products}")

print("\n--- Total Revenue per Product ---")

for product, total in revenue.items():
    print(f"{product}: ${total}")

print("\n--- Most Expensive Product per Category ---")

for category, (product, price) in
max_price_per_category.items():
    print(f"{category}: {product} (${price})")
```

Output:

--- Unique Products per Category ---

Electronics: {'Laptop', 'Monitor', 'Mouse'}

Furniture: {'Lamp', 'Desk Chair'}

--- Total Revenue per Product ---

Laptop: \$1200

Mouse: \$50

Monitor: \$300

Desk Chair: \$150

Lamp: \$45

--- Most Expensive Product per Category ---

Electronics: Laptop (\$1200)

Furniture: Desk Chair (\$150)

4. Multi-Source Log Aggregator

```
raw_logs = [
    ("10:00:01", "192.168.1.1", 200),
    ("10:00:02", "192.168.1.5", 404),
    ("10:00:02", "192.168.1.5", 404),
    ("10:00:03", "192.168.1.1", 500),
    ("10:00:04", "172.16.0.45", 403),
    ("10:00:05", "192.168.1.1", 200),
    ("10:00:06", "172.16.0.45", 403)
]

unique_logs = set(raw_logs)

error_logs = []
for timestamp, ip, status in unique_logs:
    if 400 <= status <= 599:
        error_logs.append((timestamp, ip, status))

ip_errors = {}

for timestamp, ip, status in error_logs:
    if ip not in ip_errors:
        ip_errors[ip] = []
    ip_errors[ip].append(timestamp)

attackers = set(ip_errors.keys())

print("--- Error Report (IP: Timestamps) ---")
for ip, times in ip_errors.items():
    print(f"{ip}: {times}")
```

```
print("Total Unique Attackers:", len(attackers))
print("Attacker List:", attackers)
```

Output:

--- Error Report (IP: Timestamps) ---

192.168.1.5: ['10:00:02']

172.16.0.45: ['10:00:06', '10:00:04']

192.168.1.1: ['10:00:03']

Total Unique Attackers: 3

Attacker List: {'172.16.0.45', '192.168.1.1', '192.168.1.5'}

5. Student Performance Analytics System

```
from collections import defaultdict
academic_records = [
    ("CS", "Alice", "Python", 90),
    ("Math", "Bob", "Calculus", 80),
    ("CS", "Alice", "Python", 90),
    ("CS", "Charlie", "C++", 70),
    ("Math", "Bob", "Algebra", 70),
    ("Physics", "David", "Optics", 85),
    ("CS", "Alice", "Algorithms", 95)
]

unique_records = set(academic_records)

dept_scores = defaultdict(list)
dept_students = defaultdict(set)
```

```

for dept, student, subject, score in unique_records:
    dept_scores[dept].append(score)
    dept_students[dept].add(student)

final_result = {}

for dept in dept_scores:
    total_students = len(dept_students[dept])
    avg_score = sum(dept_scores[dept]) /
len(dept_scores[dept])
    final_result[dept] = (total_students, avg_score)

print("Department | Unique Students | Avg Score")
print("-----")

for dept in sorted(final_result.keys()):
    students, avg = final_result[dept]
    print(f"{dept} | {students} | {avg}")

```

Output:

Department | Unique Students | Avg Score

CS | 2 | 85.0

Math | 1 | 75.0

Physics | 1 | 85.0